

Data Structure

Level 1

Singly Linked List Data Structures Operations

The background of the slide features a complex, abstract network of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in a discussion or collaborative work, adding a human element to the technical theme.

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Insert at Beginning



Definition

InsertAtBeginning() adds a new node at the start of the linked list:

■ Steps:

- Allocate memory for a new node
- Assign the node's value
- Point next of the new node to the current head
- Update head to point to the new node



Explanation

- ◆ The new node becomes the first node in the list.
- ◆ The previous head becomes the second node.
- ◆ This method preserves the rest of the list intact.

Procedures (Insert at Beginning)

- 1 Start with an empty linked list $\rightarrow$ Node* head = NULL
- 2 Call InsertAtBeginning(head, value)
- 3 Inside the function:
 -  Allocate memory for a new node
 -  Set new_node->value = value
 -  Link new_node->next to current head
 -  Update head = new_node
- 4 Repeat for each new element
- 5 Traverse the list to print using PrintList(head)

Example

Insert values in order: 1, 2, 3, 4, 5

➡ Resulting linked list:

5 → 4 → 3 → 2 → 1 → NULL

Note

Insertion at the beginning is $O(1)$ since it doesn't require traversal.

Warning

Ensure the head pointer is passed by reference (`Node*& head`) so the original head is updated.

Summary

- ✓ InsertAtBeginning creates a new node
- ✓ Links it to the current head
- ✓ Updates head to new node
- ✓ Resulting list shows reverse insertion order

Thank you 😊